



Wiz's Journey: Mastering the States of Matter

Producing a Serious Game to Aid in Teaching Matter and Their Phase Changes to Grade 6 Learners

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ABSTRACT

Chemistry education contains foundational concepts that are often perceived as difficult for students to understand, particularly in terms of their application to real-world situations. This difficulty may lead to cognitive overload, especially when instruction lacks opportunities for practical engagement. Interactive learning approaches allow students to actively apply and develop their analytical skills while discovering how chemistry concepts relate to everyday experiences.

This study explores the use of Game-Based Learning (GBL) through the development of *Wiz's Journey: Mastering the States of Matter*, an educational serious game designed as a learning material for Grade 6 students under the MATATAG curriculum in the Philippines. The game focuses on physical and chemical changes of matter, expanding students' understanding beyond the three common states of matter. Through interactive gameplay, students engage with concepts such as phase changes and heat transfer while learning to distinguish physical changes from chemical changes.

The primary goal of this study is to improve students' conceptual understanding, motivation, and emotional engagement by reinforcing hands-on learning and real-world applications of chemistry concepts. This project aims to provide both students and teachers with an accessible and engaging instructional resource that supports visualization, experimentation, and the development of analytical skills in basic chemistry education.

Keywords: physical and chemical changes, game-based learning, educational serious game

Sustainable Development Goals: 4: Quality Education, 11: Sustainable Cities and Communities, 12: Responsible Consumption and Production

ABSTRAK

Ang pag-aaral ng kimika ay naglalaman ng mga pundamental na konsepto na kadalasang itinuturing na mahirap unawain ng mga mag-aaral, lalo na pagdating sa aplikasyon nito sa totoong buhay. Ang ganitong kahirapan ay maaaring magdulot ng cognitive overload, partikular kung kulang ang mga aralin sa praktikal at interaktibong gawain. Sa pamamagitan ng interaktibong pagkatuto, aktibong nahahasa ng mga mag-aaral ang kanilang kasanayan sa pagsusuri at mas nauunawaan nila kung paano nagagamit ang mga konsepto ng kimika sa pang-araw-araw na buhay.

Sinisiyasat ng pag-aaral na ito ang paggamit ng Game-Based Learning (GBL) sa pamamagitan ng pagbuo ng *Wiz's Journey: Mastering the States of Matter*, isang pang-edukasyong seryosong laro na idinisenyo bilang pantulong na materyal sa pagkatuto para sa mga mag-aaral sa Baitang 6 sa ilalim ng MATATAG kurikulum sa Pilipinas. Nakatuon ang laro sa pisikal at kemikal na pagbabago ng materyal, na nagpapalawak sa kaalaman ng mga mag-aaral lampas sa tatlong karaniwang kaanyuan ng matter. Sa pamamagitan ng interaktibong gameplay, natutuhan ng mga mag-aaral ang mga konsepto tulad ng pagbabago ng estado at paglipat ng init, habang malinaw na natutukoy ang kaibahan ng pisikal at kemikal na pagbabago.

Layunin ng pag-aaral na ito na mapabuti ang pag-unawa, motibasyon, at emosyonal na pakikipag-ugnayan ng mga mag-aaral sa pamamagitan ng hands-on na pagkatuto at aplikasyon ng mga konsepto ng kimika sa mga sitwasyong hango sa totoong buhay. Nilalayon din ng proyektong ito na magsilbing kapaki-pakinabang na materyal para sa mga guro at mag-aaral upang mas mapadali ang pagkatuto ng kimika sa isang biswal, interaktibo, at makabuluhang paraan.

Mga susing salita: pisikal at kemikal na pagbabago, game-based learning, pang-edukasyon na laro

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Science education at the elementary level serves as a foundational stage in developing scientific literacy, conceptual reasoning, and problem-solving skills. Within the MATATAG Curriculum of the Department of Education (DepEd), Grade 6 Science introduces learners to fundamental concepts related to materials, including the description of changes of state for solids, liquids, and gases—specifically melting, evaporation, freezing, and condensation—using diagrams and flowcharts. These competencies aim to help learners understand observable phenomena through scientific representations and structured reasoning.

Despite the importance of these competencies, research has consistently shown that students experience difficulty in understanding the particulate nature of matter and the mechanisms underlying phase changes. Learners often memorize definitions of melting or evaporation without fully comprehending the particle-level interactions that explain these transformations. Misconceptions frequently arise regarding particle spacing, motion, and the role of heat transfer in altering states of matter (Adadan, 2013; Paik, Kim, Cho, & Park, 2004). These challenges are compounded by the abstract nature of microscopic representations, which are not directly observable.

In many classroom settings, demonstrations of phase change are limited to water-based experiments, such as melting ice or boiling water. While these activities provide observable macroscopic changes, they often lack explicit integration with particle-level models, resulting in fragmented conceptual understanding. Furthermore, access to laboratory materials for demonstrating processes such as sublimation or deposition is limited at the elementary level, restricting exposure to diverse examples of state change.

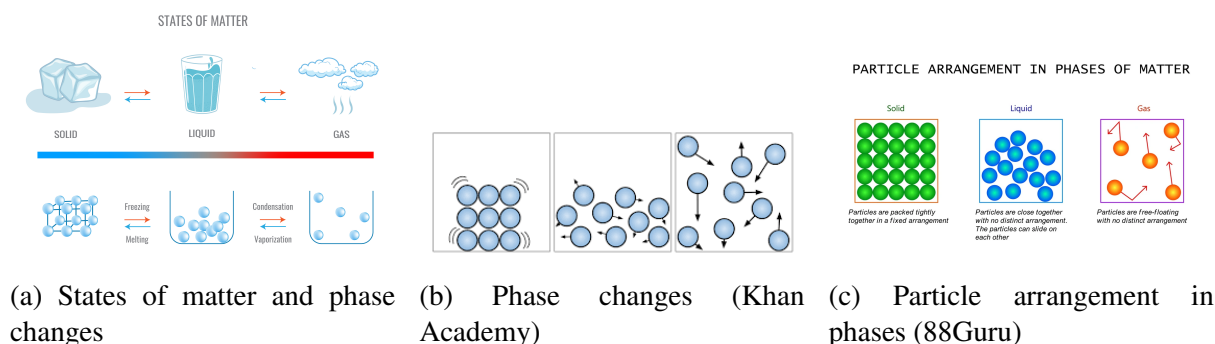


Figure 1.1: Examples of phase change diagrams used in science education

Game-Based Learning (GBL) has emerged as a promising instructional approach that integrates interactive systems with curricular objectives. Educational games can provide dynamic visualizations, immediate feedback, and experiential learning opportunities that reduce cognitive overload and promote conceptual change (Plass, Homer, & Kinzer, 2015). When designed with explicit learning outcomes and aligned assessments, digital games may enhance engagement and deepen understanding, particularly in abstract scientific domains.

This study introduces *Wiz’s Journey: Mastering the States of Matter*, a 2D educational platformer developed as a supplementary and enrichment learning tool aligned with the MATATAG Grade 6 Science competency on changes of state. The game integrates explicit terminology (e.g., “Solid → Liquid: Melting”) while allowing learners to discover appropriate phase changes through environmental problem-solving. Central to its design is an interactive particle-lattice minigame—a core mechanic of the gameplay—in which learners manipulate bonds and particle spacing to simulate state transitions. By directly engaging with representations similar to those used in classroom diagrams, learners repeatedly encounter visual models of particle behavior in a contextualized, goal-driven environment.

Given the increasing emphasis on learner-centered instruction under the MATATAG framework, there is a need to explore innovative digital tools that complement formal instruction without replacing it. This study examines whether a platformer-based educational game can support conceptual understanding, engagement, and reinforcement of curriculum-aligned competencies related to phase changes of matter.

1.2 Objectives

1.2.1 General Objective

This study aims to design, develop, and evaluate an educational platformer game that rapidly operationalizes a selected learning competency from the first quarter of the Grade 6 MATATAG Science curriculum. The game will be developed with established pedagogical and assessment strategies in mind to ensure alignment with curriculum standards and learning outcomes. Learner performance and engagement will be evaluated through pretest and posttest analyses to determine the effectiveness of the intervention.

1.2.2 Specific Objectives

Specifically, the study aims to:

1. Design learning outcome mechanics (LOMs) that strategically target both lower-order and higher-order cognitive learning outcomes aligned with the MATATAG Grade 6 Science competency on changes of state.
2. Develop a functional 2D educational platformer game that incorporates the designed LOMs and integrates interactive particle-level simulations to visually represent state transitions.
3. Evaluate the game's efficacy in teaching the selected learning outcomes to Grade 6 learners through structured pre- and post-assessment instruments, engagement surveys, and qualitative feedback from students regarding usability and perceived learning value.

1.3 Scope and Limitations of the Study

This study focuses exclusively on the development and evaluation of *Wiz's Journey: Mastering the States of Matter*, a 2D educational platformer designed to reinforce the Grade 6 Science competency from the MATATAG Curriculum Guide (CG) 2023 for Grades 3-10. Specifically, the game targets the Quarter 1: Materials learning competency that requires students to “describe changes of state for solids, liquids, and gases as melting, evaporation, freezing, and condensation using diagrams and flowcharts.” The game addresses physical changes only and does not include instruction on chemical reactions or other chemical changes.

The game mechanics are designed to reflect the chosen learning outcomes, focusing on interactive methods for handling game content that directly map to the MATATAG competency. During the development phase, which commenced in December 2025 and extended through March 2026, the development team reviewed existing educational materials and pedagogical approaches to identify optimal mechanics and instructional strategies that could be contextualized for Grade 6 students' current knowledge of phase changes.

The game was developed using Unity as the game engine, chosen for its cross-platform compatibility and the development team's proficiency with the tool. The game is deployed on multiple platforms for greater accessibility: primarily on Android mobile devices via Itch.io, as well as on web platforms via Unity Play. This multi-platform approach ensures that the game can reach students regardless of their available device.

The evaluation methodology consisted of two phases: a formative playtest involving a small sample of students in a controlled environment with qualitative observations and surveys for iterative improvements, followed by a summative playtest with a larger sample that incorporated pretest, gameplay, and posttest assessments to measure learning outcomes.

However, recruitment and participation differed from the initially planned scope. The study originally planned to recruit participants from nearby schools including Inchican Elementary School, Seven Pillars Catholic School, and Princeton Academy. In actuality, only Inchican Elementary School was able to provide informed consent and participate in the study. The school provided 5 consenting students: 3 participants for the formative playtest and 18 participants for the summative playtest (compared to the initially promised 30 students). The other targeted schools were unable to participate due to logistical constraints, including lack of response to communication attempts and the timing of the study during the end of the academic school year when students were occupied with moving-up ceremonies from Grade 6 to first-year high school.

The study does not measure long-term retention of learning outcomes or academic performance beyond the duration of the intervention. Additionally, the research evaluates the game as a supplementary enrichment tool and does not employ a control group design, limiting the ability to make direct comparisons with traditional classroom instruction. The study adopts a single-group quasi-experimental design with pre- and post-assessments, engagement surveys, and qualitative interviews focused on student perception and usability feedback.

1.4 Significance of the Study

This study contributes to elementary science education in the Philippines by examining the integration of interactive digital modeling within a platformer game environment. For learners, *Wiz's Journey: Mastering the States of Matter* provides repeated exposure to particle-level representations within meaningful problem-solving contexts, strengthening conceptual connections between diagrams and observable phenomena related to phase changes. By aligning the game with the MATATAG Grade 6 Science competency, the study demonstrates how educational games can reinforce curriculum-aligned learning objectives in accessible, engaging ways.

For educators and school administrators, this study offers insights into how educational games may complement MATATAG-aligned instruction without replacing formal teaching. The findings may inform future curriculum integration strategies, digital resource development, and professional development initiatives for teachers adopting game-based learning approaches. Given the resource constraints faced by many Philippine schools, this game's availability on multiple platforms (web and mobile) addresses accessibility barriers and demonstrates feasibility of deploying educational technology in resource-limited contexts.

For researchers and the broader educational community, this study adds to the growing body of literature on game-based learning in science education, particularly within the

Philippine educational context and addressing the K-12 curriculum transition. By examining a discovery-based yet explicitly labeled game design combined with systematic evaluation using pre-post assessments and engagement surveys, the study contributes to ongoing discussions on balancing constructivist learning principles with guided instruction in digital environments. The qualitative feedback from student participants and actual implementation challenges encountered (such as participant recruitment constraints) provide realistic insights into deploying educational games in authentic school settings.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Philippine Education Curriculum

2.1.1 Current Curriculum of Philippines Education

The Philippines' education system has undergone significant institutional changes in recent years, with the K-12 program serving as the current standardized framework. The K-12 program was implemented to align the Philippine curriculum with international standards and improve compatibility with global education systems (Abragan, Abarcas, Aquino, & Bagongon, 2022). However, recent critical assessments have revealed substantial gaps between intended learning outcomes and actual student performance in classroom settings. Research indicates that discrepancies emerge from multiple factors, including curriculum design, instructional delivery approaches, and alignment of pedagogical objectives with assessment practices (Descartin et al., 2023).

A significant challenge identified by Kilag et al. (2024) is curriculum overload—the K-12 program's extensive content requirements leave insufficient time and space for meaningful student-centered learning. In response to disruptions caused by the COVID-19 pandemic and the shift to remote instruction, the Department of Education introduced the Most Essential Learning Competencies (MELC) framework (Marasigan & Soliman, 2021). The MELC was designed to distill curricula to their core components, ensuring that the most critical competencies were prioritized during instruction (Zalun, 2023; Gallego et al., 2024). Despite these intentions, implementation has revealed persistent challenges—both new issues emerging from MELC adoption and enduring problems that preceded it (Angay, 2023; Gallego et al., 2024).

2.1.2 Status of the Philippine Education

The broader educational landscape in the Philippines reflects substantial variation in system readiness and implementation capacity across regions and school types. Urban and metropolitan schools, particularly private institutions, generally demonstrate stronger infrastructure and resource availability, enabling more robust curriculum implementation. In contrast, rural and underserved urban schools face systemic capacity constraints that compromise their ability to deliver the full breadth of competencies outlined in national curriculum frameworks (Kilag et al., 2024).

Teaching force preparedness remains a significant concern. While Filipino teachers demonstrate strong commitment and pedagogical dedication, many lack formal professional development in contemporary instructional methodologies—particularly those integrating technology, differentiated instruction, and competency-based assessment (Abragan, Abarcas, Aquino, & Bagongon, 2022). Furthermore, administrative support systems for curriculum implementation vary widely, with limited monitoring and feedback mechanisms in resource-constrained schools to ensure fidelity to curriculum standards (Gallego et al., 2024).

Student learning outcomes in foundational subjects such as science and mathematics have shown modest improvement trends but remain below regional benchmarks. International assessments and national standardized tests indicate persistent achievement gaps correlating with school resource levels and geographic location. These patterns underscore the need for targeted interventions addressing both systemic capacity and localized resource constraints (Descartin et al., 2023).

2.1.3 Challenges of Resource Poor Schools

Resource scarcity presents a fundamental barrier to equitable curriculum implementation in Philippine schools. Many schools, particularly in rural and urban marginalized communities, contend with insufficient learning materials, overcrowded classrooms, and limited access to contemporary instructional tools (Kilag et al., 2024). Abragan and Abarcas et al. (2022) documented that teachers in resource-poor schools lack adequate training in technology integration and have minimal access to information and communication technology (ICT) equipment—tools that could enhance student engagement and maintain focus during instruction.

A critical gap in science education infrastructure is the scarcity of functional science laboratories in resource-poor schools. Many institutions lack dedicated laboratory facilities, essential equipment (microscopes, measuring apparatus, safety equipment), and consumable materials required for hands-on experimentation in chemistry and physics. This absence forces teachers to rely exclusively on theoretical demonstrations and textbook illustrations, severely limiting students' ability to conduct direct observations and experiential learning in abstract scientific domains such as phase changes and matter properties (Kilag et al., 2024).

Curriculum implementation challenges are further compounded by deficiencies in teacher preparation. Curriculum unpacking—the process by which educators analyze curriculum competencies to derive specific learning objectives, instructional materials, and assessment activities—requires specialized training and pedagogical expertise. Many teachers lack formal guidance in this process, resulting in literal interpretation of competency statements without meaningful contextualization or sequencing (Gallego, Andea, et al., 2024). This insufficient preparation,

combined with limited access to resources, leads to compromised mastery of prerequisite concepts, creating foundational gaps that undermine student performance in higher-order learning objectives (Gabriel, Nepomuceno, Kadusale, Taneo, & Cabello, 2022).

Research consistently demonstrates that resource-rich schools—particularly private institutions with superior facility access and professional development infrastructure—achieve significantly higher academic outcomes than their resource-poor counterparts (Angay, 2023). To address this inequity, scholars have called for expanded provision of learning resources across all school contexts. The present study responds to this imperative by developing an educational platformer game focused on reinforcing Grade 6 competencies related to phase changes and states of matter. By leveraging the wide availability of mobile devices and web-accessible platforms, this game-based intervention can reach students regardless of whether their schools possess traditional science laboratory facilities. The game provides interactive, visual simulations of molecular-level phenomena aligned with MATATAG competencies, enabling students to engage with abstract scientific concepts through accessible gaming technology that requires minimal institutional infrastructure beyond basic internet connectivity.

2.2 Chemistry and Science Education

2.2.1 Particle Models and Phase Transitions in Elementary Learning

Understanding the particulate nature of matter and phase transitions is fundamental to chemistry and science literacy. Samarapungavan, Bryan, Staudt, and colleagues (2023) explored kindergarten and early elementary students' learning of simple particle models to explain the properties and behavior of matter in solid, liquid, and gas states during phase transitions. Their research found that when students are explicitly taught to use particle models as explanatory frameworks, they develop more accurate mental representations of microscopic processes underlying observable macroscopic phenomena. The study emphasizes that science instruction focusing solely on concrete, directly observable phenomena without explicit instruction in particulate models limits students' ability to understand abstract concepts fundamental to chemistry.

2.2.2 Misconceptions in Melting and Freezing Cycles

A critical challenge in science education is bridging the gap between macroscopic observations and microscopic explanations. Smith and Villarreal (2015) assessed 155 freshman general chemistry students' misconceptions about particle position and arrangement during the melting-freezing cycle using the Melting Cycle Instrument, which employs particulate-level

representations. Their findings revealed that many students maintained inaccurate mental models about how particles behave during phase transitions, particularly regarding particle spacing and motion during state changes. These misconceptions persist even after formal instruction, suggesting the need for innovative pedagogical approaches that make particle-level interactions more salient and comprehensible to learners across educational levels.

2.2.3 Remediation of Matter Misconceptions Through Knowledge Revision

The persistent nature of matter misconceptions has prompted researchers to investigate interventions designed to remediate student errors. Korur, Korumaz, and Avci (2025) developed the Knowledge Revision Components (KReC) framework, which integrates refutation texts and advanced organizer materials to explicitly address and eliminate misconceptions about matter and physical/chemical changes. Their research demonstrated that systematic, theory-based approaches to misconception remediation significantly improve student conceptual understanding when learners are directly confronted with the inadequacy of their existing mental models. This approach aligns with pedagogical principles emphasizing active engagement with conflicting ideas as a mechanism for conceptual change.

2.2.4 Middle School Misconceptions About Matter Cycles and Conservation

At the middle school level, Arslan (2023) employed a word association test to investigate 457 eighth-grade students' cognitive perceptions and misconceptions regarding matter cycles and environmental phenomena. The study revealed widespread misconceptions about evaporation, condensation, and the conservation of matter during state changes, particularly among students in lower-performing schools with limited access to interactive learning materials. These findings underscore how resource constraints in Philippine schools compound the challenge of teaching abstract science concepts and exacerbate existing achievement gaps. The representation and presentation of scientific concepts significantly influence student understanding, particularly when integrating multiple representational modes—text, diagrams, animations, and interactive manipulatives—to support more robust conceptual development.

2.3 Game-Based Learning

2.3.1 Developmental Strategies of Edugames

Constructivist Learning Theory asserts that knowledge is actively constructed through interaction and experience rather than passively received. Game-based environments embody constructivist principles by placing learners in active problem-solving roles. In platform-based

game environments, players experiment, test strategies, and receive immediate feedback from system responses. This aligns with the constructivist view that learning occurs through engagement and reflection upon action (Kolb, 1984; Bruner, 1966).

Outcome-based methodologies provide a structured framework for developing game-based learning environments aligned with explicit educational competencies. Sison and colleagues (2018) presented a formal Outcomes-Based Edugame Design methodology that operationalizes this alignment through a systematic process. This methodology begins by explicitly determining learning outcomes (LOs), selecting an appropriate game genre, and writing a game premise that contextualizes the learning objectives. Subsequently, the design phase proceeds incrementally through iterative “Build Game Learning Environment (GLE)” cycles, where each cycle focuses on designing a learning outcome mechanic, integrating related game elements, connecting with previously designed mechanics, and conducting playtesting and evaluation. This process repeats for each learning outcome before final integration of all elements and comprehensive evaluation (Sison et al., 2018). This approach ensures that gameplay mechanics directly support the achievement of specified educational outcomes, creating coherence between instructional goals and interactive gameplay.

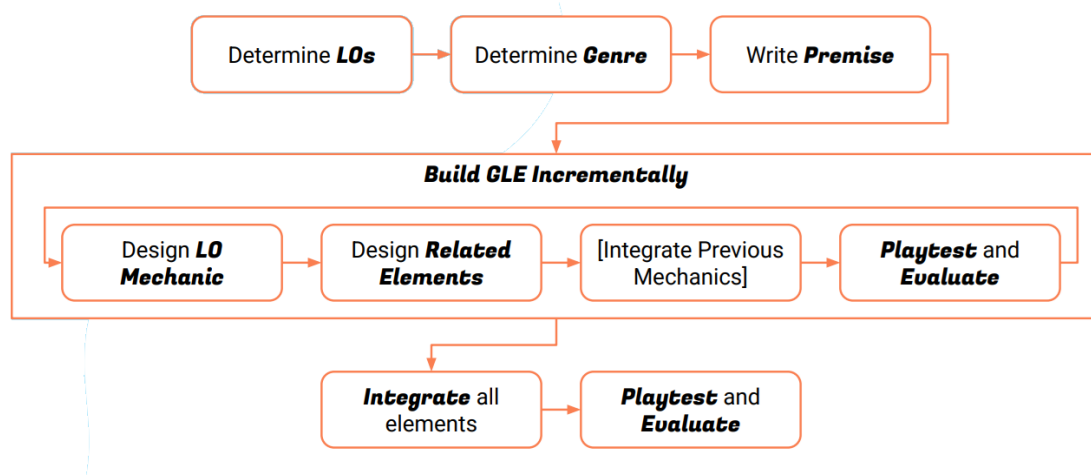


Figure 2.1: Outcomes-Based Edugame Design Methodology (Sison et al., 2018)

In game development research, iterative design is a widely accepted approach for refining gameplay mechanics, usability, and user experience (Gee, 2003). Iterative development involves repeated cycles of testing, evaluation, modification, and regression validation. Campaign testing refers to structured playthrough sessions aimed at identifying usability issues, progression bottlenecks, and gameplay friction points. Engagement evaluation models assess multiple dimensions including focused attention, satisfaction, perceived learning, and intrinsic motivation. User-centered design principles emphasize incorporating player feedback into successive iterations to improve overall game quality (Prensky, 2001). The Outcomes-Based Edugame Design methodology institutionalizes this iterative approach by embedding playtesting and evaluation at every stage of development (Sison et al., 2018), ensuring that pedagogical

alignment is maintained throughout the design process rather than evaluated only at the end. Educational game development should adopt a quasi-experimental assessment approach with pre- and post-gameplay measures to evaluate learning effectiveness, employing both quantitative metrics (test scores, learning gains) and qualitative methods (engagement surveys, observational feedback) to comprehensively evaluate user experience and educational impact.

2.3.2 Effectiveness of games in education

Multimedia Learning Theory suggests that learning is enhanced when verbal and visual information are presented in coordinated and complementary ways (Mayer, 2001, 2009). Effective multimedia design integrates text, visuals, and animation to support meaningful cognitive processing. Research specifically demonstrates that animations with textual labels significantly enhance learning outcomes in scientific domains where direct observation is limited (Mayer & Moreno, 2002). Educational games that integrate explicit textual labeling (e.g., “Solid → Liquid: Melting”) alongside dynamic visualization can support the formation of coherent mental representations of abstract processes.

Game-Based Learning Theory emphasizes the role of challenge, feedback, and goal-oriented interaction in promoting learner engagement (Prensky, 2001). Educational games combine instructional objectives with gameplay mechanics to foster sustained attention and intrinsic motivation. Engagement is a critical factor in educational technology adoption. Games that balance challenge and usability are more likely to maintain player interest and promote exploratory interaction (Gee, 2003). Assessment of engagement using structured instruments can determine whether a game builds demonstrates positive user experience.

Experiential Learning Theory posits that learning occurs through cycles of experience, reflection, conceptualization, and experimentation (Kolb, 1984). Simulated environments allow learners to experience phenomena that may not be directly observable in traditional classrooms. By simulating molecular interactions during freezing, melting, and evaporation, educational games can provide experiential exposure to abstract scientific processes aligned with curriculum standards. Outcome-Based Teaching and Learning emphasizes alignment between instructional activities and clearly defined competencies. Game level design can encourage players to apply conceptual knowledge to solve in-game challenges, aligning with higher-order cognitive objectives in frameworks such as Bloom’s Taxonomy (Anderson & Krathwohl, 2001).

2.3.3 Game-Based Learning in the Philippines

International research has demonstrated the effectiveness of educational games in enhancing engagement and conceptual understanding in science subjects. Studies evaluating interactive

simulations and digital learning games report improvements in learner motivation and positive user experience when multimedia elements and active manipulation are integrated. Additionally, research on usability evaluation models such as MEEGA+KIDS highlights the importance of assessing engagement, focused attention, satisfaction, and perceived learning in educational game contexts (von Wangenheim, Petri, & Borgatto, 2020). Studies on iterative game development emphasize that structured playtesting contributes significantly to improved playability, stability, and user satisfaction (Sison et al., 2018).

Local studies in educational technology within the Philippine context indicate increasing integration of digital learning tools aligned with curriculum standards. Research evaluating instructional games commonly employs usability surveys and engagement instruments to assess learner reception. However, limited research combines structured iterative development practices with formal engagement evaluation within educational game development projects in the Philippine context. This gap supports the relevance of developing and evaluating curriculum-aligned educational games that address the learning needs of Filipino students while demonstrating rigorous development and evaluation methodologies.

2.4 Theoretical Framework

This study is grounded in a hierarchical integration of theoretical frameworks that collectively explain both the pedagogical rationale for game-based learning and the methodological approach employed in development and evaluation.

Tier 1: Primary Theoretical Anchors

Conceptual Change Theory serves as the primary foundation, particularly given the documented misconceptions about phase changes identified in the research literature (Posner, Strike, Hewson, & Gertzog, 1982; Section 2.2). This theory posits that learners construct initial mental models of phenomena through everyday experiences; these models often conflict with scientifically accurate explanations. Conceptual change occurs not through passive information reception, but through active confrontation with the inadequacy of existing models (Posner et al., 1982). In the context of this study, students hold misconceptions about particle behavior during phase changes; the game is designed to create problem-solving scenarios that expose the limitations of these mental models and provide experiential evidence supporting more accurate particle-level explanations (Samarapungavan et al., 2023).

Constructivism complements Conceptual Change Theory by emphasizing that knowledge is actively constructed through interaction, experience, and reflection rather than passively received (Bruner, 1966; Kolb, 1984). In a constructivist game environment, learners do

not receive explicit lectures on phase changes; instead, they manipulate particle spacing and bond strength through the interactive particle-lattice minigame, discovering the relationships between these manipulations and observable state transitions. This active engagement supports the conceptual change process by anchoring abstract ideas (particle motion) to concrete, manipulable experiences within the game (Kolb, 1984).

Multimedia Learning Theory addresses how to represent abstract concepts through coordinated presentation of multiple information channels (Mayer, 2001, 2009). The game integrates text labels (e.g., “Solid to Liquid: Melting”), animated particle representations, and interactive mechanics to convey phase change concepts. Research demonstrates that coordinated text-visual-animation presentations enhance learning outcomes compared to single-modality instruction, particularly for abstract domains where direct observation is limited (Mayer & Moreno, 2002). By simultaneously showing particle lattice structures, temperature effects, and textual terminology, the game leverages multimedia principles to support learner comprehension of microscopic phenomena.

Tier 2: Supporting Theories

Game-Based Learning Theory justifies the use of challenge-based interactive mechanics to foster sustained engagement (Prensky, 2001; Gee, 2003). This theory emphasizes that games combine immediate feedback, clear goals, balanced challenge-to-skill ratios, and intrinsic motivation mechanisms. In *Wiz’s Journey*, these elements manifest as: (1) instant feedback when particles respond to player manipulations, (2) explicit win/fail conditions for phase change challenges, (3) progressive difficulty scaling, and (4) reward systems that reinforce both achievement and conceptual exploration (Prensky, 2001).

Experiential Learning Theory supports the use of simulated environments for learning abstract scientific processes (Kolb, 1984). This theory describes learning as occurring through cyclical experience-reflection-conceptualization-experimentation. The game environment allows learners to experience phase change phenomena that cannot be directly observed (molecular-level particle motion), reflect on the outcomes of their manipulations, form conceptual understanding of particle behavior, and experimentally test these understandings in subsequent challenges. This cycle repeats across levels, deepening conceptual sophistication (Kolb, 1984).

Tier 3: Operational Frameworks

Outcome-Based Teaching and Learning (OBTL) principles ensure alignment between pedagogical theory and curricular competencies (Sison et al., 2018). The game’s learning outcome

mechanics (LOMs) are explicitly designed to target specific competencies from the MATATAG Grade 6 Science curriculum, ensuring that gameplay directly supports achievement of standards-based objectives rather than generic skill development.

Bloom's Taxonomy guides cognitive level alignment (Anderson & Krathwohl, 2001). Lower-order objectives (remember, understand) are addressed through labeled particle representations and discovery mechanics; higher-order objectives (apply, analyze) are addressed through complex multi-phase challenges requiring strategic particle manipulation to solve environmental puzzles. This hierarchical alignment ensures the game scaffolds learning from foundational understanding to sophisticated application.

Iterative Design Principles and User-Centered Development Frameworks justify the two-phase approach (Section 2.5). Rather than conducting summative evaluation immediately after development, the methodology incorporates repeated cycles of playtesting, feedback integration, and refinement (Gee, 2003; Prensky, 2001). This approach recognizes that pedagogical theory informs initial design, but actual player experiences reveal implementation gaps; iterative cycles resolve these gaps before formal evaluation, ensuring the final game genuinely embodies the theoretical principles that justify its use (Sison et al., 2018). The Outcomes-Based Edugame Design methodology (Sison et al., 2018; Figure 2.1) formalizes this iterative process by requiring playtesting and evaluation after each learning outcome mechanic is designed, creating multiple feedback loops that systematically align pedagogical theory with gameplay implementation.

In synthesis, these frameworks operate hierarchically: Primary theories explain the deep pedagogical mechanisms (why games can facilitate conceptual change through constructivist interaction and multimedia learning); supporting theories specify motivational and experiential mechanisms (how games sustain engagement through challenge and experiential cycles); and operational frameworks translate these mechanisms into specific game features (which competencies to target and how to align gameplay to those targets). Iterative development ensures theoretical alignment, while structured evaluation validates achievement of alignment.

2.5 Conceptual Framework

The study adopts an expanded Input–Process–Output (IPO) model integrating both pedagogical foundations and iterative development practices. This framework visualizes how theoretical inputs are operationalized through development and evaluation processes to produce measurable outputs supporting student learning.

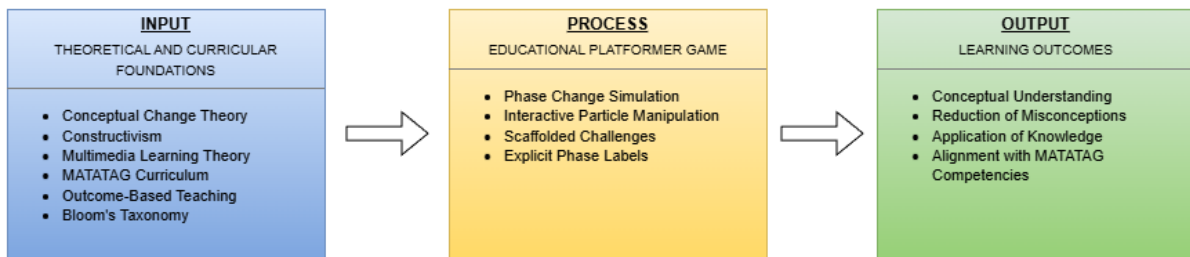


Figure 2.2: Conceptual Framework of the Study

2.5.1 Input Component

The Input component comprises three integrated elements:

1. **Theoretical Foundations:** Conceptual Change, Constructivism, Multimedia Learning, Game-Based Learning, and Experiential Learning theories (described in Section 2.4) provide the pedagogical justification for game-based intervention. These theories collectively explain why interactive gameplay can support understanding of abstract phase change phenomena when traditional instruction alone proves insufficient. The Outcomes-Based Edugame Design methodology (Sison et al., 2018) operationalizes these theoretical foundations by requiring explicit alignment of each learning outcome mechanic with underlying pedagogical principles.
2. **Curricular Alignment:** The MATATAG Grade 6 Science competency on changes of state defines the content scope. Bloom's Taxonomy and Outcome-Based Teaching principles establish cognitive level expectations and learning outcome targets. This ensures the game reinforces competencies students are expected to master rather than tangential content. The Sison et al. (2018) methodology begins with explicit determination of learning outcomes, ensuring curricular alignment from the inception of design.
3. **User-Centered Design Principles:** Iterative design, playtesting feedback loops, and engagement evaluation models (such as MEEGA+KIDS) guide the development process (von Wangenheim, Petri, & Borgatto, 2020). These principles ensure the game is not only pedagogically sound in theory but also usable, engaging, and motivating in practice. The Sison et al. (2018) methodology systematically embeds user testing through playtesting at every development stage, preventing theory-practice misalignment.

2.5.2 Process Component

The Process component is deliberately divided into two sequential phases, each with distinct purposes:

Iterative Development Phase (Internal Refinement)

- **Campaign testing:** Systematic playthrough of game levels to identify progression issues, accessibility barriers, and gameplay friction
- **Regression testing:** Verification that modifications made in response to feedback do not reintroduce previously resolved issues
- **Playtesting feedback integration:** Incorporation of player observations, difficulty reports, and suggestions into game mechanics and usability improvements
- **Gameplay refinement:** Iterative modification of particle mechanics, level difficulty curves, narrative elements, and UI design based on accumulated feedback

This phase prioritizes alignment between pedagogical theory (Inputs) and actual player experience. For example, if Constructivist principles suggest players should discover phase relationships through experimentation, but playtesting reveals players are confused by ambiguous feedback mechanisms, refinements clarify feedback without resorting to explicit instruction, preserving the constructivist approach while improving comprehension. The Outcomes-Based Edugame Design methodology (Sison et al., 2018) institutionalizes this alignment process through incremental game learning environment construction: each learning outcome mechanic is designed, integrated with related elements, connected to previously designed mechanics, and then immediately playtested and evaluated (Figure 2.1). This staged approach ensures that pedagogical alignment is validated at every step of development rather than only at the conclusion.

Final Evaluation Phase (Summative Assessment)

- **Structured play session using finalized build:** Controlled gameplay environment with Grade 6 participants
- **MEEGA+KIDS usability and engagement assessment:** Quantitative measurement of focused attention, satisfaction, perceived learning, and intrinsic motivation dimensions (von Wangenheim, Petri, & Borgatto, 2020)
- **Qualitative feedback collection:** Post-play interviews or surveys capturing student perceptions of learning value, difficulty appropriateness, and enjoyment

This phase validates whether the refined game achieves its intended educational and motivational objectives.

2.5.3 Output Component

The Output component captures both developmental and evaluative results:

Developmental Outputs

- Improved gameplay stability and reduced technical issues
- Refined user experience with reduced navigation friction and clarity-enhanced feedback mechanisms
- Optimized difficulty progression supporting both accessibility and appropriate challenge

Evaluative Outputs

- Measured engagement levels across MEEGA+KIDS dimensions indicating learner motivation and perceived learning
- Evaluated usability confirming Grade 6 students can successfully operate game mechanics without excessive frustration
- Evidence of conceptual understanding gains through pre-post assessment comparisons (evaluated in later research phases beyond this thesis scope)

The framework illustrates a systematic progression: theoretical foundations inform initial design, iterative development resolves implementation gaps, and final evaluation confirms both the game's usability and the validity of the theoretical approach. Importantly, the separation of iterative development from summative evaluation ensures that classroom-ready game builds undergo rigorous user testing before formal assessment, maximizing likelihood that measured outcomes reflect genuine intervention effectiveness rather than usability limitations masking pedagogical potential.

CHAPTER 3

RESEARCH METHODOLOGY

This chapter describes the systematic approach employed to design, develop, and evaluate *Wiz’s Journey: Mastering the States of Matter*. The methodology integrates the Outcomes-Based Edugame Design framework (Sison et al., 2018, as presented in Section 2.3.1) with structured evaluation practices aligned with the conceptual framework (Section 2.5).

3.1 Research Activities

The development of the educational game was grounded in explicit research activities that informed the design direction and ensured pedagogical alignment throughout the development process.

3.1.1 Research of Learning Outcomes and Pedagogical Approaches

Prior to commencing game design, a comprehensive research phase was undertaken to establish the pedagogical foundation for *Wiz’s Journey*. This research comprised three complementary components:

Curriculum Analysis and Learning Competency Clarification

The MATATAG Grade 6 Science curriculum documents were reviewed to identify and clarify the specific learning competencies related to phase changes and states of matter. This analysis determined the precise learning outcomes (LOs) that would anchor the game design. The target competency—understanding phase transitions (melting, freezing, evaporation, condensation) and the relationship between heat transfer and state changes—was operationalized into explicit learning outcome mechanics (LOMs) that would structure both lower-order and higher-order gameplay challenges (Sison et al., 2018).

Pedagogical Methodology Review

Existing pedagogical approaches to teaching phase changes were examined, encompassing traditional lecture-based instruction, hands-on laboratory experimentation, visual demonstrations,

and conceptual change strategies. This review informed the selection of constructivist and experiential learning principles (Section 2.4) as the theoretical foundations guiding interactive gameplay design. The review identified that effective instruction on phase changes requires students to move beyond memorization toward active conceptualization of particulate-level mechanisms underlying macroscopic observations (Samarapungavan et al., 2023).

Review of Existing Educational Resources and Games

Existing educational games, simulations, and learning aids designed to support phase change instruction were reviewed to identify design patterns, pedagogical strengths, and limitations. This comparative analysis ensured that *Wiz's Journey* would employ evidence-based design practices while offering novel approaches to addressing documented learning challenges in this domain.

3.2 Game Design Process

The design and development of *Wiz's Journey* followed the Outcomes-Based Edugame Design methodology proposed by Sison et al. (2018), which operationalizes the alignment between learning outcomes, game mechanics, and iterative user feedback. The design process proceeded through three major stages, with development occurring from late November 2025 through March 2026.

3.2.1 Stage 1: Learning Outcome Mechanics Planning and Prototyping (Late November – Mid-December 2025)

During this stage, the explicit learning outcomes derived from the curriculum analysis (Section 3.1.1) were translated into learning outcome mechanics—specific gameplay elements designed to support the achievement of each learning outcome (Sison et al., 2018). This stage involved:

- Definition of learning outcome mechanics aligned with the six primary phase changes (melting, freezing, evaporation, condensation, sublimation, deposition)
- Prototyping of interactive mechanics allowing players to manipulate variables affecting phase transitions (temperature, particle count, bond stiffness, bond length)
- Validation that each mechanic directly instantiates the underlying scientific principle it is designed to teach

- Initial playtesting of prototype mechanics to assess feasibility and pedagogical potential

This stage ensured that the foundational game mechanics were theoretically grounded before proceeding to genre and narrative development.

3.2.2 Stage 2: Genre and Premise Development (December 2025 – January 2026)

Following validation of learning outcome mechanics, the game genre and narrative premise were established to contextualize the mechanics within engaging gameplay. Decisions during this stage included:

- Selection of the 2D platformer genre as appropriate for integrating phase change mechanics with navigation challenges
- Development of the game premise: players assume the role of “Wiz,” a young wizard apprentice solving environmental puzzles involving phase transitions
- Narrative framing of each level’s challenges as requiring players to manipulate phase states to progress
- Integration of the narrative context with learning outcome mechanics to ensure mechanics served both gameplay and pedagogical objectives

This stage balanced entertainment value with pedagogical rigor, ensuring that gameplay progression was intrinsically motivated by learning objective achievement.

3.2.3 Stage 3: Iterative Building and System Integration (Late January – March 2026)

The final stage involved incremental construction of the complete game learning environment (GLE) incorporating all designed systems, mechanics, and narrative elements. Following the Sison et al. (2018) framework, this stage proceeded through multiple iterative cycles, each consisting of:

1. **Design LO Mechanic Implementation** – Addition or refinement of gameplay mechanics supporting specific learning outcomes
2. **Design Related Elements** – Development of supporting game systems (particle lattice visualization, level progression, UI, narrative scaffolding)

3. **Integrate with Previous Systems** – Connection of newly developed mechanics with previously completed systems to ensure coherent gameplay
4. **Playtesting and Evaluation** – Structured testing with participants to identify usability issues, gameplay friction, and pedagogical alignment
5. **Iteration and Refinement** – Modification based on accumulated feedback before advancing to the next learning outcome mechanic

This iterative approach, as conceptualized in the Conceptual Framework (Section 2.5), embedded continuous user testing throughout development rather than conducting comprehensive evaluation only upon completion. The embedded playtesting cycles systematically validated alignment between pedagogical theory and actual player experience, resolving implementation gaps incrementally.

By March 2026, the iterative building process yielded a complete, tested game build incorporating all designed learning outcome mechanics and supporting systems, positioning the game for formal evaluation.

3.2.4 Recruitment and Evaluation Challenges

The iterative development and evaluation phases encountered significant recruitment challenges that shaped the scope and implementation of the study. Initial contact with Inchican Elementary School was made in December 2025 but received no response during that period. Subsequent attempts to recruit Grade 6 students from Seven Pillars Catholic School and Princeton Academy were made from mid to late February 2026, but neither school responded to communication attempts. Inchican Elementary School was finally confirmed as a partner in early March 2026, providing access to a limited number of Grade 6 students willing to participate. These recruitment constraints, attributed primarily to the graduation season when students in the target grade were transitioning to secondary education and unavailable for research participation, necessitated adaptation of the planned evaluation methodology.

To address anticipated recruitment challenges, community outreach strategies were implemented preemptively during the formative evaluation phase, including public dissemination of the game through the Itch.io distribution platform and Facebook. While these platforms provided broader access to the game for feedback purposes, they did not substantially increase the number of consenting Grade 6 participants available for structured, in-person evaluation sessions.

Despite these constraints, the data gathered during the limited formative playtesting (three participants) proved valuable in amplifying signals for actual adjustments that informed

the summative evaluation design. The feedback from these three formative participants guided substantive refinements to gameplay mechanics, UI clarity, and overall user experience before the summative evaluation, ensuring that the final build represented an iteratively improved product. This adaptation demonstrated the critical value of even small-scale formative testing in validating pedagogical alignment and usability before formal evaluation.

3.2.5 Game-Based Learning Environment Evaluation

The evaluation of *Wiz's Journey* employed a mixed-methods approach combining quantitative usability and engagement measurement with qualitative player feedback. Evaluation occurred in two phases corresponding to the Process Component structure described in Section 2.5.2.

Formative Evaluation Phase (Iterative Development)

During the iterative building stage (Stage 3 above), formative evaluation data were collected to guide ongoing refinement:

- **Sample:** Three (3) Grade 6 student participants recruited through convenience sampling and institutional partnerships with Inchican Elementary School. Community outreach was conducted concurrently through Facebook and the Itch.io game distribution platform to gather feedback from a broader audience and extend reach to additional schools, though these channels did not yield sufficient consenting Grade 6 participants for in-person evaluation
- **Procedure:** Participants engaged in structured play sessions with development builds, followed by semi-structured interviews and MEEGA+KIDS-based survey feedback
- **Data Collection:** Responses were gathered via Google Forms surveys and face-to-face interviews conducted after gameplay
- **Analysis:** Feedback was qualitatively analyzed to identify usability barriers, confusion points, and suggestions for game refinement
- **Iteration:** Identified issues were prioritized and addressed in subsequent development cycles before the next formative test session

This formative phase directly informed the iterative refinement process, ensuring that the finalized build incorporated player-informed improvements.

Summative Evaluation Phase (Final Structured Evaluation)

Upon completion of iterative development, a formal evaluation of the finalized game build was conducted:

- **Sample:** Eighteen (18) Grade 6 student participants recruited through institutional partnership with Inchican Elementary School
- **Procedure and Data Collection:** Participants completed a structured evaluation session consisting of three sequential components: (1) completion of a 25-question pretest (approximately 15 minutes) assessing baseline knowledge of phase changes, (2) gameplay with the finalized build (approximately 30 minutes), and (3) completion of a 25-question posttest (approximately 15 minutes) assessing post-intervention knowledge. The posttest employed different questions from the pretest but maintained equivalent content coverage and difficulty level. Due to time constraints within the school setting, participants were unable to complete the MEEGA+KIDS survey instrument following gameplay. However, extensive qualitative evidence was gathered through systematic visual observation of participant reactions, frustrations, body language, and gameplay behaviors during the evaluation session
- **Analysis:** Quantitative analysis of learning gains employed statistical hypothesis testing. Shapiro-Wilk tests were conducted first to determine normality of the pretest-posttest difference scores. If the data were normally distributed, parametric paired t-tests were performed to assess statistically significant differences between pre- and post-intervention knowledge. If data violated normality assumptions (non-parametric distribution), Wilcoxon Signed-Rank tests were conducted instead. These tests provided statistical evidence regarding whether observed improvements between pretest and posttest scores were statistically significant, allowing for inference beyond the sample to the broader population of Grade 6 students. Qualitative observational data were analyzed thematically to identify patterns in participant engagement, frustration points, and behavioral indicators of learning or confusion

The summative phase provided empirical evidence regarding the game's achievement of the stated objectives related to usability, engagement, and learning outcomes.

3.2.6 Consent for Participation

Given that all participants were Grade 6 students (minors), rigorous informed consent procedures were implemented to ensure ethical compliance and participant protection:

- **Parental Consent and Student Assent:** A written informed consent form was developed and reviewed collaboratively with concurrent research teams engaging the same participant population. The form clearly described the study's purpose, procedures, data collection methods, confidentiality safeguards, and participants' rights. Separate assent forms were provided to Grade 6 student participants, allowing them to provide informed agreement to participate independently of parental consent. Both the parental consent forms and student assent forms were distributed through children's teachers, requiring written signatures from both guardians and students prior to any participation
- **Data Confidentiality and Participant Rights:** All forms, surveys, and study materials explicitly stated that participants could withdraw from evaluation activities at any time without penalty, that their data would remain confidential, and that data could be deleted upon request. No personally identifiable information was included in subsequent data analysis or reporting
- **Safeguarding Minors:** The requirement for both written parental consent and student assent served as primary safeguards for minor participants, respecting both parental authority and the child's own voice and autonomy in research participation. Additionally, all evaluation sessions were conducted in school settings with supervision by school personnel, and no data collection occurred without documented parental and student authorization

These procedures ensured that participation was informed, voluntary, and protective of participant rights and privacy.

3.3 Research Instruments and Data Collection

The study employed structured instruments aligned with the evaluation components outlined in the Conceptual Framework (Section 2.5).

3.3.1 MEEGA+KIDS Usability and Engagement Instrument

The MEEGA+KIDS instrument (von Wangenheim, Petri, & Borgatto, 2020) was designed to assess multiple dimensions of player experience:

- **Focused Attention:** Concentration during gameplay and ability to maintain engagement
- **Perceived Learning:** Participants' perception of learning value and knowledge gain

- **Satisfaction:** Overall enjoyment and contentment with the gaming experience
- **Usability:** Ease of understanding game mechanics and navigating game systems
- **Challenge:** Appropriateness of difficulty and balance between accessibility and cognitive demand
- **Intrinsic Motivation:** Engagement driven by enjoyment rather than external rewards

Participants responded to Likert-scale statements (typically five-point scales: Strongly Disagree to Strongly Agree) assessing each dimension. This instrument provided quantifiable measures of engagement and usability suitable for characterizing player experience.

For the purposes of this study, these six dimensions were conceptually consolidated into five primary categories to facilitate focused analysis of player experience:

- **Usability and Interface Design:** Encompasses the “Usability” dimension, assessing whether participants could effectively navigate game menus, understand on-screen prompts, and operate game mechanics without excessive confusion or frustration
- **Engagement and Experience:** Synthesizes “Focused Attention” and “Intrinsic Motivation” dimensions, measuring whether participants remained mentally engaged with the game and whether motivation was driven by inherent enjoyment rather than external pressures
- **Perceived Learning Support:** Directly corresponds to “Perceived Learning” dimension, capturing participants’ subjective assessment of whether gameplay supported their understanding of phase change concepts
- **Aesthetics and Challenge:** Combines “Satisfaction” and “Challenge” dimensions, evaluating both the visual and interactive appeal of the game and the appropriateness of difficulty progression relative to player skill levels
- **Open-Ended Feedback:** Captures participant suggestions, qualitative observations about strengths and weaknesses, and thematic patterns emerging from participant responses that the Likert-scale instrument may not capture

3.3.2 Qualitative Feedback Collection

In addition to the MEEGA+KIDS instrument, participants provided open-ended qualitative feedback through:

- **Semi-Structured Interviews:** Face-to-face conversations during and immediately following gameplay sessions, with participants answering guiding questions about strengths, challenges, confusion points, and suggestions
- **Open-Ended Questionnaires:** Written responses to prompts addressing perceived learning value, difficulty appropriateness, enjoyment factors, and recommendations for improvement

Qualitative data provided rich contextual understanding of participant experience beyond numerical measurements, enabling identification of specific usability barriers and design opportunities.

3.3.3 Visual Observation of Participant Behavior

Beyond structured survey instruments and interviews, systematic visual observation of participant behavior during gameplay sessions provided rich qualitative evidence regarding engagement, frustration, and learning processes. Throughout evaluation sessions, researcher observations documented:

- **Engagement Indicators:** Participant body language, facial expressions, and posture during gameplay (e.g., leaning forward during challenging moments, relaxed posture during familiar mechanics)
- **Frustration and Confusion Points:** Observable reactions to difficulty spikes, unclear UI elements, or unintuitive mechanics, such as hesitation, repeated failed attempts, or verbal expressions of confusion
- **Interaction Patterns:** How participants approached level challenges, whether they experimented with mechanics, sought hints, or made strategic decisions
- **Learning Indicators:** Observable changes in approach between early and later gameplay, suggesting internalization of game concepts or learning from failure
- **Social Behavior:** Interactions with researchers or peers during gameplay, including questions asked, comments made, and collaborative behaviors

These observational data were documented through researcher field notes recorded during and immediately following evaluation sessions, providing contextual richness and non-verbal evidence of participant experience that complemented quantitative and structured qualitative measures.

3.3.4 Researcher Field Notes and Observational Documentation

Throughout the iterative building phase (Stage 3), the researcher maintained detailed field notes documenting observations during playtesting sessions:

- Bug occurrence and severity observations (e.g., crashes, progression blocks, visual glitches)
- Gameplay progression issues and bottlenecks observed during participant sessions
- Player confusion points and mechanic clarity concerns noted from participant reactions
- UI navigation difficulties observed or expressed by participants
- Performance stability observations (e.g., frame rate drops, loading delays)
- Reflective session notes and developer observations regarding the alignment between intended and actual gameplay experiences

These field notes provided qualitative documentation of identified issues and observed resolutions across development iterations, enabling assessment of improvement trends and informing design refinements.

3.4 Data Analysis Methods

Analysis methods were selected to align with the research questions and evaluation objectives.

3.4.1 Quantitative Analysis of Learning Outcomes

Pretest-posttest score differences were analyzed using formal statistical hypothesis testing. First, a Shapiro-Wilk normality test was conducted to determine whether the distribution of pretest-posttest difference scores conformed to a normal distribution. Based on the results of this normality assessment:

- **If normally distributed:** A paired-samples t-test was performed to assess whether mean posttest scores differed significantly from mean pretest scores, providing evidence of statistically significant learning gains
- **If non-parametric (non-normal distribution):** A Wilcoxon Signed-Rank test was used as the non-parametric alternative to the t-test, assessing whether posttest scores were statistically significantly higher than pretest scores

These statistical tests allowed inference regarding whether observed improvements between pretest and posttest represented genuine learning differences beyond what would be expected by random variation. All quantitative analysis was conducted using appropriate statistical software with alpha level set at 0.05 for determination of statistical significance.

3.4.2 Qualitative Analysis of Observations and Feedback

Qualitative data from field notes, interviews, and open-ended questionnaire responses were analyzed using thematic coding:

- **Initial Coding:** Observational and interview data were reviewed inductively to identify emergent themes and patterns without predetermined categories
- **Theme Development:** Codes were organized into higher-order thematic categories representing distinct aspects of participant experience and identified usability concerns (e.g., “Mechanic Clarity,” “Engagement Factors,” “Difficulty Appropriateness,” “Learning Perception,” “Frustration Points”)
- **Pattern Identification:** Frequency of theme occurrence was documented to distinguish high-consensus observations from isolated concerns
- **Integration with Quantitative Data:** Qualitative themes were integrated with quantitative findings to provide comprehensive characterization of player learning and experience

3.4.3 Iterative Development Improvement Analysis

Field notes from the iterative building phase (Stage 3) and formative evaluation were analyzed descriptively to assess systematic improvements across development cycles:

- Reduction in bug frequency and severity of reported issues across iteration cycles
- Improvement in gameplay stability observations (reduction in crashes, progression blocks)
- Resolution of recurring usability concerns noted in player interactions
- Comparison of early vs. late iteration observations regarding mechanic clarity and engagement

These analyses provided evidence of systematic improvement driven by iterative user feedback and researcher observations, validating the effectiveness of the embedded evaluation and refinement approach.

3.5 Ethical Considerations

This study adhered to established ethical principles for research involving human participants, particularly minors:

- **Informed Consent:** Written parental consent was obtained prior to any participant involvement, with clear disclosure of study procedures and participant rights
- **Voluntary Participation:** Participants and parents were informed of the voluntary nature of participation and the right to withdraw without penalty
- **Confidentiality and Privacy:** No personally identifiable information was retained in data analysis or reporting. Participant responses were treated as confidential, and individual data could be deleted upon request
- **Protection of Minors:** The requirement for parental written consent, combined with evaluation in supervised school settings, provided safeguards appropriate to the minor status of participants
- **Appropriate Data Use:** All collected data were used exclusively for academic research purposes related to this thesis study

CHAPTER 4

DESIGN AND IMPLEMENTATION

CHAPTER 5

GAME SYSTEM

5.1 Software Overview

Wiz's Journey: Mastering the States of Matter is developed using Unity 2022 LTS, a cross-platform game engine selected for its extensive capabilities in 2D game development, robust documentation, and the development team's prior expertise. The game is designed as a 2D platformer that integrates educational content with interactive gameplay mechanics, allowing students to progress through levels while encountering learning outcomes related to phase changes of matter.

The game is deployed across multiple platforms to ensure widespread accessibility and reach diverse student populations. The primary deployment target is Android mobile devices, distributed through Itch.io, a popular game distribution platform that allows easy access without requiring app store approval processes. A secondary web-based version is deployed via Unity Play, allowing students to play directly in web browsers on computers and tablets without installation requirements. This multi-platform approach reflects the varying technological infrastructure available to schools and students in the Philippines and aligns with the goal of making the educational game accessible regardless of device availability.

The development timeline began in December 2025 at the conclusion of the first trimester and extended through March 2026, spanning the academic term break and into the second trimester. This timeline was established to accommodate the academic calendar constraints and allowed the development team to complete the game while managing other coursework obligations. By the time development concluded in March 2026, the target Grade 6 students had already completed their first quarter instruction on phase changes under the MATATAG curriculum, positioning the game as a supplementary learning reinforcement tool rather than a primary instructional resource.

The game architecture incorporates both lower-order and higher-order mechanics aligned with the selected learning outcomes. Lower-order mechanics focus on foundational understanding of the four phase changes—melting, freezing, evaporation, and condensation—using visual representations and simple interactions. Higher-order mechanics build upon this foundation to address more complex concepts such as the relationship between heat transfer and state changes, the particulate nature of matter during phase transitions, and critical thinking about real-world applications of phase changes. The system integrates a particle lattice visualization system to support students' conceptualization of microscopic-level processes underlying

macroscopic observations, addressing a common misconception identified in science education research.

CHAPTER 6
RECOMMENDATIONS